

Dominique G. Maucieri, Ph.D., M.Sc.

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University
of Victoria



UNIVERSITY OF
CALGARY

Education

- 2021 – 2025 **Doctor of Philosophy** in Marine Biology, Biological Sciences, University of Victoria, BC, Canada, Senior Advisor: Dr. Amanda Bates
Thesis: *Heat, food availability and habitat degradation: How environmental conditions shape the resilience and physiological responses of marine invertebrates across scales.*
- 2019 – 2021 **Master of Science** in Marine Biology, Biological Sciences, University of Victoria, BC, Canada, Senior Advisor: Dr. Julia K. Baum
Thesis: *Impacts of local and global stressors on coral biodiversity.*
- 2014 – 2019 **Bachelor of Science (Honours)** in Ecology and Zoology, Biological Sciences, University of Calgary, AB, Canada
Ecology Honours Advisor: Dr. Robert M.R. Barclay
Zoology Honours Advisor: Dr. Jessica M. Theodor
Awarded First Class Honours for GPA and Research Excellence
Ecology Thesis: *Consumption of spiders by Albertan bats.*
Zoology Thesis: *Sexual dimorphism in bat wing morphology – variation among foraging styles*

Publications (ORCID: 0000-0003-1849-2472)

1. **Maucieri, D.G.**, D.J. Kushner, and A.E. Bates. Enduring a major marine heatwave: The role of local cool refugia and kelp forests in the resilience of marine invertebrates. *Diversity and Distributions* 31(6):1-13. DOI: <https://doi.org/10.1111/ddi.70044>
2. Starko, S., G. Epstein, L. Chalifour, K. Bruce*, D. Buzzoni*, M. Csordas*, S. Dimoff*, R. Hansen*, **D.G. Maucieri***, J. McHenry*, K.L. Tietjen*, B. Timmer* and J.K. Baum. 2025. Ecological responses to extreme climatic events: a systematic review of the 2014-2016 Northeast Pacific marine heatwave. *Oceanography and Marine Biology - An Annual Review*. 63. *Author order alphabetical
3. Shaffer, J. A., Oxborrow, B., Parks, D. S., **Maucieri, D.G.**, & Michel, J. 2024. Linking Marine Ecosystem Response to Shoreline Armor Removal and Large Dam Removals in the Elwha River and Nearshore, Washington, USA. *Journal of Coastal Research*. 113: 890-895.
4. **Maucieri, D.G.**, S. Starko, and J.K. Baum. 2023. Tipping points and interactive effects of chronic human disturbance and acute heat stress on coral diversity. *Proceedings of the Royal Society B: Biological Sciences* 290:20230209. DOI: [10.1098/rspb.2023.0209](https://doi.org/10.1098/rspb.2023.0209).
5. Baum, J.K., D.C. Claar, K.L. Tietjen, J.M.T. Magel, **D.G. Maucieri**, K.M. Cobb, and J.M. McDevitt-Irwin. 2023. Transformation of coral communities subjected to an unprecedented heatwave is modulated by local disturbance. *Science Advances* 9(14):1-14. DOI: [10.1126/sciadv.abq5615](https://doi.org/10.1126/sciadv.abq5615).
6. **Maucieri, D.G.** 2022. quadcleanR: An R Package for the Cleanup and Visualization of Quadrat Data. *Journal of Open Source Software* 7(79): 4674. DOI: [10.21105/joss.04674](https://doi.org/10.21105/joss.04674).
7. **Maucieri, D.G.**, and J.K. Baum. 2021. Impacts of heat stress on soft corals, an overlooked and highly vulnerable component of coral reef ecosystems, at a central equatorial Pacific atoll. *Biological Conservation* 262: 1-10. DOI: [10.1016/j.biocon.2021.109328](https://doi.org/10.1016/j.biocon.2021.109328)

8. **Maucieri, D.G.**, A.J. Ashbaugh, and J.M. Theodor. 2021. Sexual dimorphism in bat wing morphology – variation among foraging styles. *Canadian Journal of Zoology* 99(11): 953-960. DOI: [10.1139/cjz-2021-0035](https://doi.org/10.1139/cjz-2021-0035)
9. **Maucieri, D.G.**, and R.M.R. Barclay. 2021. Consumption of spiders by the little brown bat (*Myotis lucifugus*) and the long-eared bat (*Myotis evotis*) in the Rocky Mountains of Alberta, Canada. *Canadian Journal of Zoology*. 99(3): 221-226. DOI: [10.1139/cjz-2020-0160](https://doi.org/10.1139/cjz-2020-0160)

In Review

1. Bruce, K.A., J.H.R. Burns, **D.G. Maucieri**, K. Pascoe, J.K. Baum. Prolonged heatwave-induced mass coral mortality influences the structural complexity of reef habitats. *In Review at Journal of Animal Ecology*.
2. Hansen, R.L.G., A. Halford, **D.G. Maucieri**, and J.K. Baum. Spatial variation in benthic community composition on a minimally disturbed coral reef after a prolonged marine heatwave. *In Review at Coral Reefs*.
3. Jones, A.E., **D.G. Maucieri**, J.M. Schuster, T.J. Bird, and A.E. Bates. Do fishers follow the rules? Evaluating recreational fishing compliance of protected areas and downstream ecological responses. *In Review at Aquatic Conservation: Marine and Freshwater Ecosystems*

Peer Review

Publons/Web of Science Researcher ID: AAA-6115-2022

- Global Ecology and Biography (2 Reviews)
- Journal of Integrative Biology (3 Reviews)
- Biological Conservation (3 Reviews)
- Oecologia (2 Reviews)
- Conservation Biology (1 Review)
- Acta Chiropterologica (1 Review)

Research Experience

- **Post-Doctoral Research Associate at the Leverhulme Centre for Anthropocene Biodiversity (LCAB)** (2026-2029), leading analytical research on biodiversity change in the Anthropocene. I evaluated how methodological choices, data types, and scale influence discrepancies in biodiversity trends, and conducted large-scale analyses of ecological datasets to understand biological responses to human-driven environmental change. Responsibilities included developing research plans, performing analytical modelling, collaborating across transdisciplinary teams, and disseminating findings through publications, conferences, and outreach activities.
- **Contract Coder with the United Nations Population Fund** (2023 – 2024), developing small area estimation models to analyze family planning indicators within Sub-Saharan African countries. I generated heatmaps aggregated to regional levels that visualized models to support future decision-making on access to family planning.
- **Intern at the South African Shark Conservancy** (2019), contributing to research focused on monitoring the local endemic catshark population and outreach with local schools. Tasks included shark measuring and tagging, analysis of baited remote underwater video (BRUV) footage, quantitative analysis of recapture data, and dive surveys.

- **Research Assistant at the University of Calgary** (2019) in Dr. Robert Barclay's Lab, tasks including the capture of bats using mist nets and harp traps, morphometric measurements, fecal analysis, and the mentorship and training of undergraduate researchers.
- **Research Assistant at the University of Calgary** (2017) in Dr. Leland Jackson's Lab, and with the Advancing Canadian Wastewater Assets (ACWA) project at the Pine Creek Wastewater Treatment Plant. Key responsibilities included conducting chlorophyll pigment extractions and ash-free dry mass analyses on biofilm collections, performing water and depth measurements in small southern Alberta lakes, and conducting comprehensive water sample analyses, including assessments of total suspended solids, chlorophyll content, phosphorus content, and zooplankton density/community composition.
- **Research Assistant at the University of British Columbia** (2016) within Dr. Kathy Martin's Lab on Hudson's Bay Mountain in Smithers, BC. Responsibilities included banding and measuring nestlings, conducting predation experiments, observing bird population behaviour, and retrieving last season's geo-locators, plus deploying new geo-locators. This work focused on the impact of predation on the development of Horned Lark (*Eremophila alpestris*) nestlings.

Teaching and Advising

Course Instruction

2024	Applied Data Analysis , Bamfield Marine Sciences Center, 24 students, Fall Program 2024
2024	Directed Studies , Bamfield Marine Sciences Center, 24 students, Fall Program 2024

Curriculum Development

2023 & 2024	Marine Ecology . Assisted in planning field trips, assignments, and quizzes as the teaching assistant for this course at Bamfield Marine Sciences Center.
2022	Study Design and Data Analysis . Assisted in creating lab R assignments and quizzes as the teaching assistant for this course at Bamfield Marine Sciences Center.
2019 – 2020	Frontiers in Marine Biology . Employed as a course developer where I designed labs and assignments, including six R tutorials and two R Assignments.

Teaching Assistant

While working as a teaching assistant, I have fulfilled a multifaceted role which has included delivering prelab presentations, facilitating student projects, grading assignments and papers, demonstrating lab procedures and experiments and facilitating discussions. Additionally, I have been able to mentor my students in their academic careers.

- **Marine Ecology**, Bamfield Marine Sciences Center, 21 students (Summer 2024)
- **Study Design and Data Analysis**, University of Victoria, approx. 45 students (Spring 2021 to 2024).
- **Marine Ecology**, Bamfield Marine Sciences Center, 22 students (Summer 2023)
- **Frontiers in Marine Biology**, University of Victoria, 29 students (Spring 2021)
- **Intro to Ecology**, University of Victoria, approx. 45 students (Fall 2019 to 2021)
- **Principle of Genetics**, University of Victoria, approx. 45 students (Spring 2020)
- **Energy Flow in Biological Systems**, University of Calgary, approx. 25 students (Fall 2018)

Guest Lecturer

Marine Ecology, BMSC, Summer 2024, 1 lecture (1.5 hours)

Marine Ecology, University of Victoria, Spring 2024, 1 lecture (50 min)

Study Design and Data Analysis, University of Victoria, Spring 2024, 1 lecture (50 min)

Survey of Invertebrates, University of Victoria, Fall 2023, 1 lecture (50 min)

Marine Ecology, BMSC Summer 2023, 1 lecture (2 hours)

Community & Ecosystem Ecology, University of Victoria, Spring 2023, 3 lectures (2.5 Hours)

Advising Experience

- 2024 – 2025 **Alec Jones**, University of Victoria (Honours, Supervisor: Dr Amanda E. Bates).
- 2023 – 2025 **Ashley Mickens**, University of Victoria (MSc Student, Supervisor: Dr Rana El-Sabaawi).
- 2023 – 2025 **Agnes Djioua**, University of Victoria (Honours, Supervisor: Dr Amanda E. Bates & Women in Science Mentorship Program).
- 2017 – 2024 **Bamfield Marine Science Center Ambassador**, University of Calgary (2017-2019) and University of Victoria (2019 – present).
- 2023 **Nerissa Kassis**, University of Victoria (Directed Studies, Supervisor: Dr Amanda E. Bates).
- 2022 – 2023 **Rebecca Krohman**, University of Victoria (Honours, Supervisor: Dr Amanda E. Bates).
- 2020 – 2021 **Rebecca Hansen**, University of Victoria (Honours, Supervisor: Dr. Julia K. Baum).
- 2022 – 2023 **Caity Sayers**, University of Victoria (Women in Science Mentorship Program).
- 2019 – 2020 **Maggie Brenneman**, University of Victoria (Women in Science Mentorship Program).
- 2018 – 2019 **Natalie Dong**, University of Calgary (Science Mentorship Program).
- 2017 – 2018 **Teagan Fulton**, University of Calgary (Science Mentorship Program).
- 2017 – 2018 **Sustainability Ambassador** with the Sustainability Resource Center, University of Calgary
- 2018 **Advisor**, Faculty of Science Research Nights, University of Calgary.
- 2018 **Peer Mentor**, University of Calgary (Energy Flow in Biological Systems, approx. 450 students)

Facilitated Workshops and Training

- Mar 2025 **Introduction to R**. Offered at the University of Victoria in collaboration with the Biology Undergraduate Society (in person, 2hrs).
- Sep 2023 **Introduction to R**. Offered at the University of Victoria in collaboration with the Biology Undergraduate Society (in person, 1.5hrs).
- July 2023 **Map Making in R**. Offered at BMSC during the Marine Ecology course (in person, 1hr)
- Apr 2022 **How to create an R Package**. Offered at the University of Victoria (virtual, 1hr).
- Summer 2020 **Coral Identification Training**, University of Victoria, four undergraduate students. (weekly 1hr meeting for two months)

Pedagogical Training

Graduate Teaching Assistant Workshop, University of Victoria, 2019

Graduate Teaching Assistant Workshop, University of Calgary, 2018

Communication and Culture (CMCL 507) at the University of Calgary, 2017

Conferences

- Jul 2025 **Maucieri, D.G.**, R.D. Stuart-Smith, G.J. Edgar, and A.E. Bates. *The energetic cost of warming: Predicting fish consumption and invertebrate abundance under ocean warming*. – 10-minute talk, Australian Marine Sciences Association (AMSA) in Melbourne, Australia.
- Jun 2025 **Maucieri, D.G.**, R.D. Stuart-Smith, G.J. Edgar, and A.E. Bates. *The energetic cost of warming: Predicting fish consumption and invertebrate abundance under ocean warming*. – Poster, International Congress for Conservation Biology (ICCB) in Brisbane, Australia.

Jun 2024	Maucieri, D.G. , D. Kushner, and A.E. Bates. <i>Enduring a major heatwave: the role of kelp forests and cold refugia in promoting marine invertebrate community resilience</i> . – 12-minute talk, North American Congress for Conservation Biology in Vancouver, BC
Jun 2023	Maucieri, D.G. , and A.E. Bates.* <i>Impacts of Multiple Stressors on Temperate Marine Invertebrate Communities and the Role of Local Refugia in Community Resilience</i> . – 12-minute talk, Coastal Zone Canada in Victoria BC
Mar 2022	Maucieri, D.G. , and A.E. Bates. <i>Impacts of multiple stressors on temperate marine invertebrate communities and the role of local refugia in community resilience</i> – 3-minute talk, Pacific Ecology and Evolution Conference in Victoria BC
Aug 2021	Maucieri, D.G. , Samuel Starko, and J.K. Baum. <i>Disentangling the impacts of multiple stressors on coral reef diversity</i> – 12-minute talk, Canadian Society for Ecology and Evolution.
July 2021	Maucieri, D.G. , Samuel Starko, and J.K. Baum. <i>Impacts of local and global stressors on the structure and beta diversity of coral communities</i> - poster presentation, International Coral Reef Symposium.
Nov 2020	Maucieri, D.G. <i>Impacts of heat stress on soft corals, an overlooked, but fundamental component of coral reef ecosystems</i> - 5-minute talk, Western Society of Naturalists Conference, https://youtu.be/tVeY2Er2ljk
July 2020	Maucieri, D.G. <i>Impacts of local and global stressors on the structure and diversity of coral reef communities</i> – 15-minute talk, Global Coral Reef Week, https://youtu.be/0MgfzmiCZoE
Feb 2019	Maucieri, D.G. <i>Sexual dimorphism in bat wing morphology – variation among foraging styles</i> – 12-minute talk, Prairie University Biology Symposium in Winnipeg, Canada.
Oct 2018	Maucieri, D.G. <i>Spider consumption by bats</i> – poster presentation, North American Symposium for Bat Research in Puerto Vallarta, Mexico.

*presenting author

Grants, Fellowships and Awards

PhD (Total: \$173,570)

2025	Dr. Arne H. Lane Graduate Fellowships in Marine Sciences (\$10,105)
2024	Student International Activities Fund (\$500)
2024	W. Gordon Fields Memorial Fellowship (\$1,575)
2024	NSERC Michael Smith Foreign Study Supplement (\$6,000)
2024	Fisheries and Oceans Canada Aquatic Science Supplement (\$10,000)
2024	Bamfield Marine Science Center Teaching Assistant Scholarship (\$2,000)
2022 – 2025	NSERC Canada Graduate Scholarship–Doctoral (CGS D; \$110,000)
2022 – 2024	President’s Research Scholarship (\$15,000)
2023	Bamfield Marine Science Center Teaching Assistant Scholarship (\$2,000)
2023	Fisheries and Oceans Canada Aquatic Science Supplement (\$10,000)
2022	Randy Baker Memorial Scholarship (\$390)
2021	VNHS Alice M. Hay Scholarship (\$1,000)

MSc (Total: \$41,949)

2021	Most viewed paper for the Canadian Journal of Zoology (DOI: 10.1139/cjz-2020-0160)
2020 – 2021	University of Victoria Fellowship (\$15,000)
2019 – 2020	NSERC CGS-M (\$23,333)
2020	Advanced Dive Training Grant - Professional (\$1,000)
2020	The Maureen De Burgh Memorial Scholarship (\$2,616)

BSc (Total: \$35,900)

2014 – 2018	Dean’s List
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2017 & 2018	Hisao P. & Mary Needler Arai Marine Field Studies Scholarship (\$2,500)
2018	PURE award (\$4,500)
2018	NSERC USRA award (awarded but not accepted due to PURE also being received; \$4,500)
2017	Bamfield Tuition Scholarship (\$700)
2017	Upland Birds Alberta Scholarship (\$1,000)
2016 & 2017	Louise McKinney Scholarship (\$5,000)
2015 & 2017	University of Calgary Undergraduate Merit Awards (\$3,000)
2016	University of Calgary General Undergraduate Scholarship (\$5,000)
2016	Grace LeBel Bursary (\$900)
2015	Jason Lang Scholarship (\$1,000)
2015	President's Admission Scholarship (\$2,000)
2014 & 2015	Gerald Roberts Mortimer and Victor Emanuel Mortimer Scholarship (\$2,500)
2014	Royal Canadian Legion (Branch 288) Scholarship (\$800)
2014	Alexander Rutherford Scholarships (\$2,500)

Professional Service

Organizing Committees

2024	Abstract Committee Chair for the University of Victoria Graduate Student Symposium
2021 – 2022	Registration and Communications Chair for the Pacific Ecology and Evolution Conference 2022. https://peec2022.wordpress.com/
2022	Social Committee member for the University of Victoria Graduate Student Symposium
2019	Abstract Committee member for the University of Victoria Graduate Student Symposium

Educational Outreach

2025	Elementary school outreach, speaking to students at A.E. Bowers Elementary School (Gr 4)
2025	Scientist at World Ocean Day Celebration hosted by the Shaw Center for the Salish Sea
2022 – 2025	Webmaster for UVic Women in Science
2021 – 2025	Judge for the Annual BC/Yukon Regional Science Fair (Annually)
2023 – 2024	Graduate Student Society co-representative for Biology, University of Victoria.
2023 – 2024	Graduate Student Society Finance Committee member, University of Victoria.
2022 – 2024	Participating Scientist with Letters To A Pre-Scientist (1 pen pal per academic year)
2019 – 2024	Participating Scientist with Skype a Scientist program, 11 classes total ranging from kindergarten to 12 th grade
2019 – 2022	Oceaneer with the Shaw Center for the Salish Seas.
2022	High school outreach, speaking to students at École Georges P. Vanier (Gr 9) and Bert Church High School (Gr 12)
2022	3-minute thesis competitor.
2019	Participating Scientist with with the South African Shark Conservancy, educating the public about shark biology and conservation.
2019	Bat expert with the Alberta Community Bat Program
2018 – 2019	Elementary school presentations on the diversity and biology of bats.

Research Facilitation

2022 – 2024	Ecstats Meetings, University of Victoria. Ecstats, is a graduate-run group at UVic that was established in 2012. Since fall 2022, I have been leading weekly Ecstats tutorials where I teach graduate student peers coding in R and work with them through statistical analyses on ecological data.
2023 – 2024	Animal Care Committee Graduate Student Representative, University of Victoria.

2021 – 2022	Lab meeting coordinator for the Bates lab
2018	Researcher for the Alberta Community Bat Program Bat Blitz in Castle Provincial Park
2016 – 2017	Volunteer with the Calgary Wildlife Rehabilitation Society

Media

2024	Interview with UVic Women in Science
2024	Interviewed for Anthro-Scene Podcast. https://open.spotify.com/episode/7u6PC3lehKuYhGXHqR5oJL?si=zs-eFqHtSuOrBZoEZvUvIQ
2023	Interview with Oceanbites Out Loud Podcast. https://oceanbitesorg.podbean.com/e/an-interview-with-dominique-maucieri/
2021	Interviewed on Impostor Syndrome for the Graduate Matters blog at University Affairs. https://www.universityaffairs.ca/career-advice/graduate-matters/you-arent-the-only-one-suffering-from-imposter-syndrome/

Software

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Coding Languages

R – Intermediate to advanced proficiency (>9 years of experience)
 SQL – Beginner proficiency (~1 year of experience)
 HTML – Beginner proficiency (~2 year of experience)
 Python – Beginner proficiency (~1 year of experience)

Professional Certifications

- Canadian Association for Underwater Science Scientific Diver (CAUS Level I)
- PADI Rescue Diver and Dry Suit Diver
 - Nitrox & Peak Performance Buoyancy specialties
- Dive Accident Management (DAM)
- DAN Oxygen Provider
- Standard First Aid and CPR-C with AED (SFA) - Lifesaving Society
- Pleasure Craft Operators Card (PCOC)
- Small Vessel Operator Proficiency (SVOP)
- Small Domestic Vessel Basic Safety (SDV-BS)
- Restricted Operator Certificate – Maritime (ROC-M)

References

Dr. Amanda Bates

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